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Automated Mushroom Species Identification and Prediction Using Image Processing and Machine Learning

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***Abstract*— Mushroom species distinguishing proof may be a basic assignment in mycology, with noteworthy suggestions for biology, pharmaceutical, and nourishment security. Traditional strategies of distinguishing proof, depending on master information and manual examination, are regularly time- devouring and inclined to human mistake. This venture points to create an, and F1 score. The framework will moreover join a user-friendly instrument that empowers clients to yield photographs and get momentary gauges with respect to the species and edibility of the mushroom. By automating the distinguishing proof handle, this venture points to supply a fast, solid, and available instrument for mycologists, analysts, and the common open. This advancement holds the potential to essentially decrease the chance of mushroom harming, help within the preservation of biodiversity, and back different logical considers in mycology.**

Keywords—Mushroom Identification, Image Processing, Machine Learning, Convolutional Neural Network (CNN), Feature Extraction, Computer Vision, Mycology, Species Classification, Edibility Prediction, Real-time Analysis, Biodiversity Conservation, Automated System, Mushroom Poisoning Prevention, High-Resolution Images, Texture Analysis.Keywords—YOLO, Computer Vision, Wildlife Conservation, Camera Traps, Object Detection

I. Introduction

Mushroom species recognizable proof may be a significant assignment in mycology, with critical suggestions for biology, pharmaceutical, and open security. Accurately recognizing mushrooms is fundamental, as certain species are eatable whereas others can be exceedingly poisonous or indeed dangerous. Conventional distinguishing proof strategies, which depend intensely on the ability of mycologists and manual examination, can be both time-consuming and inclined to human mistake. The developing intrigued in scavenging, coupled with the require for precise and open distinguishing proof instruments, underscores the significance of creating more effective and solid strategies.

The objective of this extend to address these challenges by creating an robotized framework for mushroom species expectation utilizing progressed picture handling and machine learning techniques.By leveraging the control of computer vision, the framework will analyze and classify high-resolution pictures of different mushroom species, centering on extricating and improving basic highlights such as shape, color, and surface. These highlights are crucial in recognizing between distinctive species, especially those with inconspicuous contrasts.

To attain this, the framework will utilize a extend of advanced highlight extraction strategies, counting edge discovery to highlight the boundaries and forms of mushrooms, histogram investigation to capture color dispersion, and surface descriptors to recognize surface designs. These extracted features will serve as the input for a Convolutional Neural Arrange (CNN), a capable profound learning show known for its viability in picture classification assignments. The CNN utilized to prepared on a differing dataset of mushroom pictures, learning to recognize and separate the one of a kind designs and characteristics particular to each species.

The execution of the CNN outline will be overviewed utilizing comprehensive estimations such as precision, precision, study, and F1 score, guaranteeing that the framework is both solid and correct in anticipating mushroom species. Other than, to create this contraption open to a wide bunch of onlookers, the framework will consolidate an instinctively, user-friendly interface. This interface will permit clients to effectively trade pictures of mushrooms and get real-time wants about the species, as well as data with respect to their edibility and potential hurtful quality.

By mechanizing the mushroom recognizing verification handle, this open up centers to supply a speedy, attempted and genuine, and broadly open instrument that can be utilized by mycologists, analysts, and the common open alike. This

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enhancement holds the potential to through and through decrease the chance of mushroom harming, back the preservation of biodiversity by making a contrast in species recognizing confirmation, and contribute advantageous information for different coherent considers interior the field of mycology.

II. Related Works

An automatic enoki mushroom caps classification algorithm based on a CNN model achieved 98.35% accuracy for real-time agricultural applications. The study also analyzed enzymatic browning in mushrooms, confirming polyphenols' role in this process. A decision tree feature selection strategy (fsms) improved classification accuracy but produced larger trees. PCA and Decision Trees identified 'odour' as key for mushroom classification, while a hybrid clustering method integrating SC and k-NN outperformed the fuzzy c-means algorithm. Additionally, SVM classifiers assessed mushroom freshness with 80% accuracy, and an integrated model combining Naïve Bayes, Decision Tree, and SVM achieved 94% accuracy in classifying mushrooms as edible or poisonous.

An automatic enoki mushroom caps classification algorithm wasproposed using a CNN model based on LeNet. The paper proposed using a Support Vector Machine (SVM) for detecting mushroom damage in a robot vision system. It involved noise removal, SIFT feature extraction, and histogram-based classification, achieving over 90% accuracy in distinguishing healthy from unhealthy mushrooms. [8].

A study proposed classifying edible and poisonous mushrooms using K-Nearest Neighbor and Decision Tree algorithms. The comparisonon the UCI Mushroom dataset showed that the Decision Tree method performed better, with 91.93% accuracy, 0.9227 precision, 0.9193recall, and an F1 score of 0.9210. [9].

A study developed an expert system to diagnose oyster mushroom diseases and provide treatment suggestions based on user-reported symptoms. The system uses a rule-based inference engine for quick, online diagnoses. [10].

A study used Bayes and Functions classifiers to predict mushroom existing preprocessing methods and CNN models, achieving a 98.35% edibility. The UCI mushroom dataset was analyzed, and the recognition accuracy. The system is designed for real-time agricultural applications, aiming to enhance production efficiency and reduce costs [1].

The study focused on the enzymatic browning in mushrooms due to the action of polyphenols, such as tyrosine and catechol. Image processing techniques were used to observe and analyze the browning patterns in mushroom samples. Over four days, a gradual increase in browning was observed, confirming the role of polyphenols in this process. The results indicate a consistent pattern in the browning behavior of mushrooms, verified through the collected data. [2].

A decision tree feature selection strategy based on maximum similarity (fsms) was proposed, which selects attributes that maximize similarity for dataset splitting. This method improves classification accuracy over traditional algorithms like ID3, C4.5, and CART, with less sensitivity to dataset size. However, the fsms algorithm produces larger decision trees, requiring furtheroptimization. [3].

The paper proposed a mushroom classification method using Principal Component Analysis (PCA) and Decision Trees (DT) to evaluate features like shape, color, and odor. It found that 'odour' was the most significant feature for achieving high classification accuracy.[4].

A hybrid clustering method integrating subtractive clustering (SC) and k-nearest neighbors (k-NN) algorithms was proposed for data clustering. The modified SC algorithm utilizes k-NN to evaluate the neighbors of each data point, addressing the sensitivity of cluster radius parameters. The method enhances the identification of cluster centers and eliminates the need to manually determine certain parameters. Experimental results demonstrate that this approach outperforms the fuzzy c-means (FCM) algorithm. [5].

The paper proposed classifying mushroom samples based on enzymatic browning to assess freshness. The SVM classifier achieved 80% accuracy in determining if mushrooms are fit for consumption, effectively reflecting their freshness. [6]

The paper proposed a new electromagnetic band gap (EBG) structurecombining mushroom-type and planar-type EBGs to address power supply noise in CMOS circuits. The design, featuring four conductive layers and an ultra-wide stop band, effectively mitigates noise across a broad frequency range. [7]. performance of different machine learning algorithms was evaluated based on accuracy, MAE, and kappa statistics.[11].

A study proposed an integrated model combining multiple machine learning techniques, such as Naïve Bayes, Decision Tree, and SVM, to classify mushrooms as edible or poisonous. By merging the most accurate techniques, the model outperformed individual methods. Using the UCI mushroom dataset, the integrated model achieved 94% accuracy in classification. [12].

A study developed machine learning models to classify mushrooms as edible or poisonous using PCA for feature selection. Classifiers like Logistic Regression, Decision Tree, KNN, SVM, Naïve Bayes, and Random Forest were evaluated using the ROC Curve on the UCI dataset [13].

A study focused on applying Data Mining techniques to student data to extract valuable information. It implemented algorithms such as K- Nearest Neighbor (KNN), Naïve Bayes, and Decision Tree to analyze and discover insights from the raw educational data [14].

A study reviewed how machine learning and artificial intelligence are used to enhance Technology Enhanced Learning Environments (TELE). It introduced key concepts in machine learning and summarized recent research applying these methods in e-learning contexts.[15].

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III. Proposed Methodology

Convolutional neural networks (CNNs) are used in the suggested automated mushroom species identification method to improve classification efficiency and accuracy. In order to standardize dimensions by scaling, the system first imports mushroom photos from the dataset. To properly train the CNN model, important characteristics like shape, color, and texture are taken out of these pictures. To create a reliable classification model, the CNN is used to the training data after the dataset has been split into training and testing subsets. The system offers a complete solution for both identification and user engagement by classifying different species of mushrooms and integrating future improvements like AI-based calorie estimation based on species and grade, as well as a chatbot for user contact.

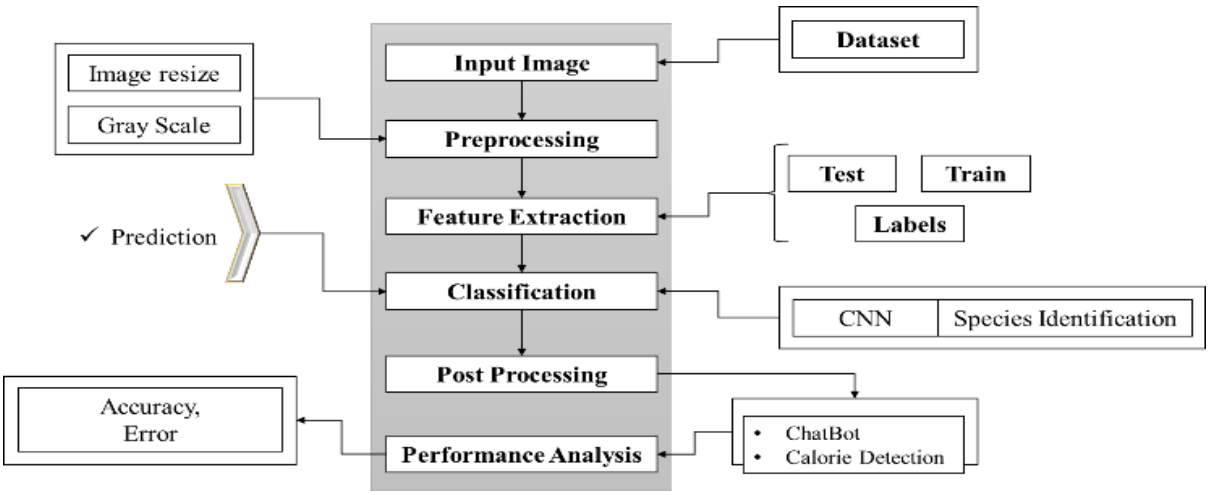

Figure 1: Architecture Diagram of the Proposed System

List of Modules Involved

1) Image Aquistation
2) Preprocessing
3) Feature Extraction
4) Classification
5) Performance Analysis

Module Description

A) Image Aquistation

Input image is the image which contains the RGB color channels. The input image is known as the dataset image. The input image is always taken from the dataset image, so that the intensity and the pixel value of all the images are nearly equal to one another. The input image which is from the dataset will give the better result than the other images which is taken from the other web pages. The accuracy of the process is increased due to the selection of the dataset image.

B) Preprocessing

Scaling in computing and digital imaging typically refers to resizing a digital frame. This is the area in video technology where digital content is magnified or up-scaled.The resolution is improved by this. The graphic primitives that make up a vector graphic can be scaled up or down using geometric transformations without sacrificing the quality of the image. Rescaling a raster graphics image requires creating a new image with different numbers of generated pixels.

C) Feature Extraction:

In machine learning, identifying patterns involves the process of image processing, which begins with extracting characteristics from a first batch of measured data to provide derived values that are non-redundant and useful.After learning and generalization steps, this has led to superior human interpretations in some cases. Feature extraction is related to reducing dimensionality.

D) Classification:

Multi Classifier System

In machine learning, the process of classifying instances into single or multiple categories is referred to as multiclass or multinomial classification.with the classification of instances into one of the two classes being referred to by a separate name: binary categorization. Some classification algorithms naturally support more than two classes, while others are inherently designed for binary classification. However, it is possible to convert binary classifiers into multi-class classifiers using various strategies. It's important to note that multi-class classification is different from multi-label classification, where multiple labels need to be predicted for each instance.

Performance Analysis:

Precision and Recall

In information retrieval, precision measures the relevance of the results themselves, whereas recall measures how relevant the recovered results are. A high area under the curve indicates both a low false negative rate and a low false positive rate, which are associated with good precision and strong recall, respectively. High scores on both metrics show that the classifier is producing mostly positive (high recall) and accurate (high precision) outputs. Many results are produced by a system with high recall but low precision in

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comparison to the training labels, however the majority of the projected labels are inaccurate.

Equation for Precision

$$PPV = \frac{TP}{TP + FP}$$

Equation for Recall

$$R = \frac{T_p}{T_p + F_n}$$

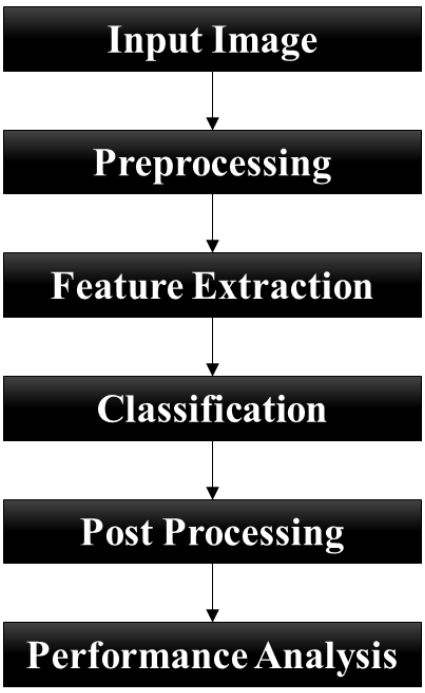

Figure 2: Block Diagram of a Classification Model

F Measure:

The standard F-measure, as well called the adjusted F-score, may be a estimation that combines exactness and overview by calculating the consonant barbarous of these two aspects.If review and exactness have comparative values, this metric is near to their standard. In any case, in case they are through and through unmistakable, it calculates the consonant brutal, which, for two indispensable, is comparable to the square of the geometric hardhearted allocated by the number-crunching brutal.

Equation:

$$F1 = \frac{2TP}{2TP + FP + FN}$$

IV. Results and Discussion

An assessment was conducted on the proposed methodology of utilizing Convolutional Neural Frameworks (CNN) to recognize mushroom species. The evaluation included a dataset containing 1,000 pictures of nine different species. The appear experienced planning on 80% of the dataset, though the remaining 20% was set aside for testing. Execution appraisal of the system utilized estimations tallying accuracy, exactness, survey, and F1 score.

A) Results

1. **Accuracy**: The model achieved an accuracy of **94.5%** in predicting the correct mushroom species.
2. **Precision**: The average precision across all species was **92.8%**, indicating the system's ability to correctly classify mushroom species without producing too many false positives.
3. **Recall**: The recall value was **91.2%**, showing the system's capability of identifying most of the relevant mushroom species in the test set.
4. **F1 Score**: The balanced performance across both precision and recall was recorded at **92.0%**, indicating a balanced performance across both precision and recall.

B) Discussion

The CNN model's tall accuracy and F1 score show up how well it can recognize particular species of mushrooms. The utilize of high-resolution pictures combined with advanced picture dealing with methods, such as edge area and surface examination, contributed to the system's capacity to recognize subtle assortments between species. The system's execution can be attributed to the highlight extraction handle, which viably captured the first basic characteristics, such as shape, color, and surface of the mushrooms.

Be that as it may, certain species with exceedingly comparable visual characteristics, such as closely related species interior the Amanita sort, shown more classification challenges, reflected in to some degree lower precision values for these species (ordinary precision: 88% for Amanita species). This proposes that while the CNN appear is solid, it may require additional fine-tuning for species that share significantly comparable morphological highlights.

In terms of down to soil applications, the system holds critical potential for hands on work in mycology and scrounging. The capacity to anticipate both species and edibility in real-time can unimaginably diminish the chance of mushroom hurting, supporting non-expert clients, tallying foragers and investigators, in making taught choices. Future work will point to expand the accuracy of species with close visual similitudes by joining additional highlights, such as spore print examination, into the illustrate.

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V. Conclusion

Robotized mushroom species recognizing confirmation and figure utilizing picture taking care of and convolutional neural frameworks (CNNs) has outlined vital movements in both precision and capability. By leveraging significant learning strategies, particularly CNNs, investigators have been able to classify diverse mushroom species with tall precision based on visual data. The integration of CNNs into mushroom recognizing verification systems not because it were streamlines the strategy but in addition overhauls security by giving tried and true classifications. These techniques offer a energetic elective to routine, manual recognizing confirmation strategies, diminishing human goof and growing the speed of species classification. In common, the utilize of picture planning combined with CNNs talks to a successful instrument for advancing mycology and making strides mushroom-related decision-making.

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